

Macro Projects:

Unit I – Algorithm Analysis (3 Projects)

1. **Sorting Complexity Visualizer** *Prompt:* "Generate a graph comparing time complexity of Merge, and Quick Sort for $n = 10, 100, 1000$." *Outcome:* Line chart showing growth rates.
2. **Recursion Tree for Merge Sort** *Prompt:* "Draw a recursion tree for Merge Sort dividing an array of 8 elements." *Outcome:* Tree showing recursive splits and merges.
3. **Asymptotic Growth Comparison** *Prompt:* "Visualize Big-O, Omega, and Theta notations with examples of algorithms." *Outcome:* Concept map linking growth functions.

Unit II – Divide and Conquer / Greedy (3 Projects)

4. **Greedy Job Sequencing Flowchart** *Prompt:* "Create a flow diagram showing job selection based on deadlines and profits." *Outcome:* Decision path visualization.
5. **Divide-and-Conquer Matrix Multiplication** *Prompt:* "Illustrate recursive division of matrices in Strassen's algorithm." *Outcome:* Block diagram of submatrix operations.
6. **Optimal Merge Pattern Tree** *Prompt:* "Draw a binary tree showing optimal merge pattern for files of sizes 10, 20, 30, 40." *Outcome:* Weighted merge tree visualization.

Unit III – Dynamic Programming (3 Projects)

7. **Traveling Salesperson DP Table** *Prompt:* "Visualize DP state transitions for TSP with 4 cities." *Outcome:* Table or graph showing subset transitions.
8. **All-Pairs Shortest Path Matrix Update** *Prompt:* "Show iterative updates of distance matrix in Floyd-Warshall algorithm." *Outcome:* Stepwise matrix visualization.
9. **Knapsack DP Table** *Prompt:* "Draw a DP table for 0/1 Knapsack with 4 items and capacity 10." *Outcome:* Grid showing item inclusion decisions.

Unit IV – Backtracking (3 Projects)

10. **N-Queens State Space Tree** *Prompt:* "Visualize the state space tree for $N = 4$ queens, marking invalid branches in red." *Outcome:* Tree showing recursive exploration.
11. **Sum of Subsets Tree** *Prompt:* "Draw the state space tree for subset sum = 15 from {5, 10, 12}." *Outcome:* Branching diagram showing inclusion/exclusion.
12. **Graph Coloring Backtracking Flow** *Prompt:* "Illustrate backtracking steps for coloring a 4-vertex graph using 3 colors." *Outcome:* Flowchart showing color assignments and backtracks.

Unit V – Branch and Bound (2 Projects)

13. **Knapsack Branch-and-Bound Tree** *Prompt:* "Illustrate bounding and pruning in 0/1 Knapsack using node values." *Outcome:* Tree with bound values and pruned branches.
14. **TSP Branch-and-Bound Search Tree** *Prompt:* "Draw a search tree showing bounding and pruning for TSP with 4 cities." *Outcome:* Hierarchical tree showing cost bounds.

Dear Students,

I am sharing the **DAA Macro Project List and detailed instructions** for your reference. Please go through the project topics and guidelines carefully.

Instructions to be Followed:

- Choose **ONE project from each Unit** (Total: **5 projects**).
- Write the **algorithm/pseudocode** for each selected project.
- Use **AI tools** to generate the required visualizations/diagrams.
- Submit both the **prompt used** and the **AI-generated diagram/visualization**.
- Provide a **short explanation of the algorithm logic and visualization**.
- Create a **GitHub repository** and upload your project files.
- Maintain a professional **README.md** for each project.
- Prepare a **PPT** summarizing your projects.
- Be prepared to **present your project in class or submit a short video walkthrough**.

Submission Deadline: 5th October 2026

I will conduct the **Macro Project Review on 5th October 2026**. Therefore, all students must ensure that their projects, GitHub repository, README files, PPT, and required visualizations are completed and ready for review by the deadline.

Please follow the **recommended repository structure and README template** shared along with the project list.

Each group must select ONE project from EACH of the five units, making a total of 5 projects per group. Each group member should work on one project from one unit. Project topics should not be repeated within the same group. The project allocation for each group will be as specified by the faculty.

Faculty GitHub Project Board:

<https://github.com/users/jhansirenuka0129-byte/projects/1/views/1>

The above Project Board will be used to monitor the progress and review status of each group.

Important: Each group must maintain ONE repository for all 5 projects. Do not create separate repositories for each project.

Thank you.

Example Integration

For Unit IV – Backtracking (N-Queens):

- Write the algorithm (pseudocode).
- Create a prompt:

“Visualize the state space tree for N=4 queens, marking invalid branches in red.”

- Generate the image.
- Explain how backtracking is shown visually.

Group	Unit I	Unit II	Unit III	Unit IV	Unit V
G1	1	4	7	10	13
G2	2	5	8	11	14
G3	3	6	9	12	13
G4	1	5	9	10	14
G5	2	6	7	11	13
G6	3	4	8	12	14
G7	1	6	8	10	13
G8	2	4	7	12	14
G9	3	5	9	11	13
G10	1	5	7	12	14
G11	2	6	8	10	13
G12	3	4	9	11	14
G13	1	6	9	10	13
G14	2	4	7	11	14

Recommended Repository Structure

```
DAA-Macro-Project-Group01/
|
|—— README.md
|
|—— Unit1_AlgorithmAnalysis/
|   |—— Project1_SortingComplexity.py
|   |—— Prompt.txt
|   |—— Visualization.png
|   |—— README.md
|
|—— Unit2_DivideAndConquer_Greedy/
|   |—— Project4_JobSequencing.cpp
|   |—— Prompt.txt
|   |—— Visualization.png
|   |—— README.md
|
|—— Unit3_DynamicProgramming/
|   |—— Project7_TSP_DP.py
|   |—— Prompt.txt
|   |—— Visualization.png
|   |—— README.md
|
|—— Unit4_Backtracking/
|   |—— Project10_NQueens.java
|   |—— Prompt.txt
|   |—— Visualization.png
|   |—— README.md
|
|—— Unit5_BranchAndBound/
|   |—— Project13_Knapsack_BnB.py
|   |—— Prompt.txt
|   |—— Visualization.png
|   |—— README.md
|
|—— PPT/
|   |—— Group01_DAA_Macro_Project.pptx
```

README Template for Each Project

Project Title

N-Queens Problem using Backtracking

Description

This project implements the N-Queens problem using backtracking and visualizes the state space tree through prompt engineering.

Algorithm

Include pseudocode or flowchart here.

Prompt Used

“Visualize the state space tree for N=4 queens, marking invalid branches in red.”

Output

Attach generated image or link.

Learning Outcome

- Understood recursive backtracking.
- Learned prompt-based visualization.
- Practiced GitHub documentation.